

Liver Transplant, Pediatric RRG

RRG: P-795-RRG (ISC)
Link to Codes

MCG Health
Inpatient & Surgical
Care
28th Edition

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Clinical Indications for Procedure and Care

- Procedure is indicated for **ALL** of the following:
 - ☐ Transplant needed for **1 or more** of the following:
 - ☐ Acute liver failure (also known as "fulminant liver failure"), as indicated by **ALL** of the following:
 - Acute liver disease (ie, 8 weeks' duration or less)^[A]
 - Hepatic encephalopathy of any degree^[B]
 - Severe manifestations of liver disease, as indicated by **1 or more** of the following:
 - Acute need for mechanical ventilation
 - Acute need for renal replacement therapy
 - International normalized ratio (INR) of 2.0 or more
 - ☐ Severe chronic liver disease, as indicated by **1 or more** of the following:
 - Model for End-Stage Liver Disease (MELD) or Pediatric Model for End-Stage Liver Disease (PELD) score of 15 or higher^{[C][D][E][F]}
 - ☐ Complication of liver disease, as indicated by **1 or more** of the following:
 - Severe growth failure (eg, z score is less than 2 standard deviations below mean for age and sex)
 - Parenteral nutrition necessary to maintain growth or euglycemia
 - Ascites or hepatic hydrothorax that persists or recurs despite diuresis and low-sodium diet
 - Hepatic encephalopathy that persists despite treatment (eg, lactulose, rifaximin)
 - Recurrent spontaneous bacterial peritonitis
 - Recurrent severe cholangitis (eg, 2 or more episodes within 6 months requiring intravenous antibiotics for 2 weeks via central access)
 - History of life-threatening infection (eg, bacteremia, hypotension) with need for ICU stay
 - Variceal bleeding that persists despite treatment (eg, beta-blocker therapy, band ligation)
 - Biliary atresia
 - Hepatocellular carcinoma with no evidence of tumor outside the liver
 - ☐ Cholangiocarcinoma and **ALL** of the following:
 - Tumor less than or equal to 3 cm in diameter
 - Unresectable due to location (eg, perihilar, vital vessel involvement), extent (eg, multifocal, bilobar disease), or severe underlying hepatic disease
 - No metastatic or nodal disease
 - Neoadjuvant chemoradiation planned prior to transplant
 - Hepatoblastoma without metastases that is not amenable to hepatic resection
 - Epithelioid hemangioendothelioma unresponsive to therapy that is not amenable to hepatic resection
 - ☐ Neuroendocrine tumor and **ALL** of the following:
 - Primary tumor and extrahepatic disease resected with curative intent 6 months or more prior to transplant
 - Stable disease for 6 months or more prior to transplant
 - Tumor not amenable to resection
 - No evidence of extrahepatic metastases
 - ☐ Polycystic liver disease and **ALL** of the following:
 - Surgical therapy (removal of cysts, fenestration) not effective or not possible
 - Severe disease, as indicated by **1 or more** of the following:
 - Hepatic decompensation (eg, ascites, hepatic encephalopathy)
 - Concurrent hemodialysis or GFR less than 20 mL/min/1.73m² (0.33 mL/sec/1.73m²)  eGFR - Pediatric Calculator
 - Prior kidney transplant
 - Moderate to severe protein calorie malnutrition
 - Severe sarcopenia

- ☐ Primary sclerosing cholangitis and **ALL** of the following:
 - Cirrhosis on imaging or liver biopsy
 - Transplant necessary due to **1 or more** of the following:
 - Biliary tract stricture not responsive to intervention
 - Intractable pruritus
 - 2 or more episodes of cholangitis
- Familial amyloidosis
- Urea cycle disorder
- Organic acidemia
- ☐ Hepatopulmonary syndrome^[G] and **ALL** of the following:
 - Sequelae of portal hypertension (eg, ascites, varices, splenomegaly, thrombocytopenia)
 - Arterial partial pressure of oxygen of less than 60 mm Hg (8.0 kPa) (on room air) not otherwise explained by underlying pulmonary disease
 - Right-to-left cardiac shunt demonstrated by contrast echocardiogram ("bubble study") or nuclear perfusion lung scan
- ☐ Portopulmonary hypertension and **ALL** of the following:
 - Portal hypertension
 - Pulmonary hypertension on right heart catheterization with **ALL** of the following:
 - Mean pulmonary artery pressure greater than 35 mm Hg at rest
 - Pulmonary vascular resistance 3 Wood units (240 dynes*sec/cm⁵) or greater
 - Other causes of pulmonary hypertension not a significant contributor to pulmonary hypertension (eg, heart failure, valvular heart disease, parenchymal pulmonary disease)
 - Satisfactory response to pulmonary hypertension treatment (eg, vasodilators, endothelial receptor blocker), as evidenced by **1 or more** of the following^[H]:
 - Post-treatment mean pulmonary artery pressure is less than 35 mm Hg and pulmonary vascular resistance is less than 5 Wood units (400 dynes*sec/cm⁵).
 - Post-treatment mean pulmonary artery pressure is more than 35 mm Hg and less than 45 mm Hg and pulmonary vascular resistance is less than 3 Wood units (240 dynes*sec/cm⁵).
- Acute decompensated Wilson disease
- Anhepatic state^[I]
- Acute hepatic porphyria with recurrent intractable symptoms despite pharmacotherapy (eg, prophylactic hemin or givosiran therapy)
- Failure of previous liver transplant
- Hepatic artery thrombosis in transplanted liver
- Other hepatic disorder associated with increased mortality that can be addressed with liver transplant (eg, glycogen storage disease, familial intrahepatic cholestasis, severe trauma, alpha-1 antitrypsin deficiency)^[E]
- No contraindications to liver transplant present

See Liver Transplant, Pediatric [ISC](#) Optimal Recovery Guideline for complete guideline and further supporting content such as Supplemental Medicare Criteria (ie, CMS' Two-Midnight Rule exceptions), Alternatives to Admission or Procedure, Optimal Recovery Course, and Evidence Summary.

Operative Status Assignment

Goal Length of Stay: 10 days postoperative

In some situations procedures that might otherwise be performed on an ambulatory basis may require inpatient care. In some cases this need can be identified preoperatively; in other cases this need is due to postoperative events or clinical findings (eg, failure to meet discharge readiness milestones). See Discharge Readiness or Extended Stay section in this guideline, or Ambulatory Surgery Exception Criteria [GRG](#), as appropriate.

- **Operative Status Criteria: Inpatient**

Level of Care Criteria

For patient conditions requiring more than routine inpatient care, documentation of need may be done using the **[Level of Care]** chart.

Intensive Care	Intermediate Care	Telemetry Care
• ICU admission may be indicated when need is demonstrated by 1 or more of the following: ☐ Vital sign abnormalities, including 1 or more of the following^[J]: ■ Systolic arterial pressure less than 90 mm Hg, or 20 mm Hg below patient's usual pressure in adult or child 10 years or older ■ Systolic arterial blood pressure less than 60 mm Hg in neonate (less than 1 month), 70 mm Hg in infant (1 month to	• Intermediate care admission may be indicated after intensive care treatment (eg, as "step-down" care), or to provide higher level of care than general hospital ward (eg, as "step-up" care in absence of intensive care admission needs) (see Intensive Care Guidelines ISC), as indicated by 1 or more of the following:	• Telemetry care admission^{[T][U]} may be indicated for 1 or more of the following: ☐ Suspected or proven acute ischemic cardiac event, including 1 or more of the following: ■ Acute coronary syndrome ■ Post-acute MI ■ Suspected vasospastic angina (to establish

1 year of age), or less than the sum of 70 mm Hg plus twice the patient's age in child 2 to 9 years of age

- Mean arterial pressure less than 65 mm Hg in adult or child 10 years or older^[K]
- Mean arterial pressure^[K] drop of greater than 30 mm Hg from baseline attributable to cardiovascular dysfunction (eg, cardiogenic shock)
- Pulse less than 40 or greater than 140 beats per minute in adult or adolescent 16 years or older
- Pulse less than 70 or greater than 150 beats per minute in child 2 to 15 years of age
- Pulse less than 90 or greater than 160 beats per minute in infant younger than 2 years

☐ Laboratory findings (new), including **1 or more** of the following:

- Pulse oximetry or arterial oxygen saturation less than 90%, or partial pressure of oxygen less than 60 mm Hg (8.0 kPa) despite oxygen supplementation^[L]
- Rising partial pressure of carbon dioxide with acute or uncompensated respiratory acidosis
- Arterial pH less than 7.2 or greater than 7.65
- Serum sodium less than 110 mEq/L (mmol/L) or greater than 160 mEq/L (mmol/L)
- Serum sodium less than 125 mEq/L (mmol/L) with Altered mental status, seizure, or respiratory arrest
- Serum sodium more than 145 mEq/L (mmol/L) with Altered mental status, seizure, diabetes insipidus requiring close monitoring of fluid balance and resuscitation, or severe dehydration requiring large-volume fluid resuscitation
- Serum potassium less than 2 mEq/L (mmol/L) or greater than 7 mEq/L (mmol/L)
- Serum potassium less than 2.5 mEq/L (mmol/L) with severe clinical or electrocardiogram manifestations (eg, paralysis, cardiac arrhythmia, delayed depolarization, flat T waves, or U waves)
- Serum potassium greater than 6 mEq/L (mmol/L) with electrocardiogram changes of hyperkalemia (eg, peaked T waves, widened QRS, cardiac arrhythmia; for patient with permanent pacemaker: widened paced QRS, increased pacemaker stimulus to QRS interval, or failure to capture)
- Calcium^[M] greater than 14 mg/dL (3.5 mmol/L) or ionized calcium greater than 10.0 mg/dL (2.50 mmol/L)
- Calcium^[M] greater than 12 mg/dL (3.0 mmol/L) or ionized calcium greater than 8.0 mg/dL (2.0 mmol/L) with Altered mental status, severe volume depletion requiring large-volume IV fluid resuscitation, Hypotension, significant arrhythmia, heart block, or digitalis toxicity

☐ Intermediate level monitoring or care needed, as indicated by **1 or more** of the following:

- Care requiring nurse-to-patient ratio of 1 to 2 or 1 to 3 (eg, complex wound management, postoperative high-risk patient, profound neuromuscular weakness)
- Frequent (but less frequent than hourly) monitoring, such as vital signs, vascular, or neurologic checks
- Stable chronic mechanical ventilator or long-term weaning needs
- Noninvasive ventilation^[R]
- High-flow oxygen therapy^[S]
- Peritoneal dialysis
- Frequent pulmonary therapy with endotracheal suctioning (nonintubated or chronic tracheostomy patient)
- Rapid diuresis for fluid overload
- Electrolyte, metabolic, or other problem requiring frequent laboratory testing or treatment adjustments
- Monitoring of continuous high-level epidural anesthesia
- Monitoring after toxic ingestion with risk for neurologic, cardiac, or respiratory compromise
- Titration of IV vasodilator or antiarrhythmic agents
- Low-dose inotropic or vasodilator therapy without need for frequent titration
- Continuous pulse oximetry monitoring
- Short-term electroencephalographic monitoring
- Patient who requires short-term inpatient monitoring (eg, cardiac, wound site, perfusion)

☐ **General Surgery** diagnoses or procedures, including patient with postoperative Hemodynamic stability requiring **1 or more** of the following:

- Close monitoring for first postoperative day (eg, due to risk of procedure, underlying comorbidity, need for frequent blood

diagnosis with ST monitoring)

- Suspected MI (until it is ruled out)
- Newly diagnosed left main coronary artery lesion (until revascularization)

☐ Cardiac rhythm disorder or conduction abnormality, including **1 or more** of the following:

- Dangerous arrhythmia diagnosed or suspected (until controlled or diagnosed as not requiring intervention)
- New-onset or recurrent atrial tachyarrhythmia
- Wolff-Parkinson-White syndrome with rapid conduction (until controlled by antiarrhythmic agent or ablation)
- Symptomatic Bradycardia^[V]
- Chronic atrial fibrillation with medical condition that may affect ventricular rate (eg, autonomic dysfunction)^[W]
- High-risk ECG findings (eg, bifascicular block, abnormal QT interval, ventricular pre-excitation, other conduction abnormality)
- Firing of implantable cardioverter-defibrillator^[X]
- Suspected pacemaker or implantable cardioverter-defibrillator malfunction^[Y]

☐ New administration or adjustment of treatment affecting cardiac rhythm or conduction, including **1 or more** of the following:

- New administration or adjustment of antiarrhythmic drug^[Z]
- New administration or adjustment of rate control therapy for atrial tachyarrhythmia
- New administration or adjustment of negative chronotropic agent (eg, beta-blocker) in patient with baseline Bradycardia
- New administration or adjustment of QTc prolonging medications^[AA]
- New wearable (external) cardioverter-defibrillator for indication anticipated to be temporary (eg, myocarditis, untreated heart failure meeting ICD criteria but expected to improve with therapy, indication for ICD with temporary contraindication to implantation, bridge to

■ Serum phosphorus less than 1 mg/dL (0.32 mmol/L) ■ Toxic drug level or poisoning causing or likely to cause neurologic abnormalities or Hemodynamic instability ■ Less severe laboratory abnormalities contributing to 1 or more of the following: • Seizure • Altered mental status • Muscle weakness or severe spasms • Arrhythmias • Hemodynamic instability • Other significant clinical manifestations ☐ Electrocardiogram (or cardiac monitoring) findings, including 1 or more of the following: ■ Inherently unstable or life-threatening arrhythmia (eg, sustained ventricular tachycardia, ventricular fibrillation, asystole) ■ Arrhythmia causing Hypotension (eg, Bradycardia, Tachycardia) ■ Complete heart block causing Hypotension ■ Other findings indicative of need for intensive care (eg, acute cardiac ischemia, myocardial infarction) ☐ Physical findings, including 1 or more of the following: ■ Threatened airway ■ Altered mental status that is severe or persistent ■ Repeated or prolonged seizures ■ New-onset anuria (urine output less than 0.1 mL/kg/hour over 4 hours) ■ Cyanosis (new) ■ Cardiac tamponade ■ Status post respiratory or cardiac arrest ■ Findings consistent with abdominal emergency (eg, peritoneal signs) ☐ Imaging findings, such as dissecting aneurysm or ruptured viscus ☐ Severe burns, as indicated by 1 or more of the following^[N] ■ Hemodynamic instability ■ Hypotension or requirement for aggressive fluid resuscitation ■ Inhalation injury ■ Respiratory insufficiency with requirement for high-flow oxygen, mechanical ventilation, or extracorporeal membrane oxygenation ■ Need for intubation or mechanical ventilation due to high risk of airway obstruction (eg, laryngeal injury seen on fiberoptic laryngoscopy, oropharyngeal burn, soot in nares or oropharynx, deep circular neck burn, stridor, change in voice, or facial burn with extensive burn involving total body surface area of 40% or greater in adult or 60% or greater in child) ■ Need for intubation or mechanical ventilation due to reduced level of consciousness (eg, Glasgow coma scale less than 8)  GCS Calculator	testing, surgical or anesthetic complication, or high postoperative risk of complications) ■ Fluid resuscitation because of major fluid shifts ■ Extensive wound management ■ Complex multiple injury care ■ Need for postoperative noninvasive ventilation	transplant or other advanced heart failure therapy) ☐ Following cardiac or major thoracic procedure, including 1 or more of the following: ■ Cardiac surgery (first 48 to 72 hours unless complications occur) ■ Noncardiac thoracic surgery (eg, pulmonary lobectomy; first 48 to 72 hours unless complications occur) ■ Nonurgent percutaneous coronary intervention with complications or suboptimal results (for at least 24 hours or until complication is resolved) ■ Short-term monitoring after cardiac procedure, as indicated by 1 or more of the following^[BB]: • Electrophysiologic studies • Implantable cardioverter-defibrillator placement • Nonurgent percutaneous coronary intervention without complications until sheath is pulled • Temporary pacemaker placement (until removed or permanent pacer placed) • Permanent pacemaker placement • Arrhythmia ablation (up to 24 hours for complex ablations (eg, pulmonary vein isolation), atrioventricular nodal ablations, or serious comorbidities) • Implantation of ventricular assist device • Transcatheter procedures (eg, aortic valve replacement, valvuloplasty, atrial or ventricular septal defect closure) • Percutaneous left atrial appendage
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- Requirement for frequent or intensive debridement and dressing changes
- Serious chemical burn
- High-voltage (eg, 1000 volts or more) electrical burn
- Carbon monoxide poisoning
- Cyanide toxicity
- Infant younger than 1 year old
- Multiple comorbidities
- Severe infection or sepsis (eg, skin and soft tissue, pneumonia, urinary infection)
- Life-threatening cardiac, renal, pulmonary, or neurologic dysfunction
- Concomitant trauma, organ failure, arrhythmia, or other medical condition requiring ICU care

☐ Specific intervention or monitoring needed, as indicated by **1 or more** of the following:

- New need for assisted ventilation, invasive or noninvasive[O]
- New need for intubation (eg, to protect airway)
- New tracheostomy (less than 48 hours old)
- Hourly vital signs or neurologic checks
- Pulmonary artery line monitoring needed
- Continuous arterial line monitoring needed
- Continuous IV vasoactive medications
- Continuous IV antiarrhythmics
- Large volume IV fluid resuscitation (eg, greater than 6 L per day)
- Large or rapid transfusion needs (eg, more than 6 units within 24 hours)
- High-risk IV treatment, such as thrombolysis, hypertonic saline, or mannitol infusion
- Rapid desensitization or graded drug provocation challenge for high-risk hypersensitivity reaction to required medication (eg, penicillin, aspirin, monoclonal antibody)
- Acute cardiac pacing
- Intra-aortic balloon pump
- Ventricular assist device
- Extracorporeal membrane oxygenation device
- Pericardiocentesis
- Hemodialysis in unstable patient
- Continuous renal replacement therapy (eg, continuous venovenous hemodialysis)
- Continuous fluid removal via hemofiltration (eg, continuous venovenous hemofiltration)
- Peritoneal dialysis initiation
- Emergency bronchoscopic therapy (eg, for hemoptysis)
- Emergency endoscopic therapy for bleeding
- Balloon tamponade for variceal bleeding
- Intracranial pressure monitoring or tissue oxygen monitoring
- Ventriculostomy monitoring
- Treatment of ongoing seizures
- Induced hypothermia or coma
- Ongoing frequent testing and treatment for acute conditions, including **1 or more** of the following:

closure (6 hours if performed under general anesthesia, shorter duration if performed under moderate sedation)

☐ Cardiac disease, including **1 or more** of the following:

- Acute myocarditis or pericarditis
- Acute decompensated heart failure (if myocardial ischemia or arrhythmia suspected etiology)
- Takotsubo cardiomyopathy (aka stress cardiomyopathy or apical ballooning syndrome)
- Infectious endocarditis
- Patient admitted with ventricular assist device (for cardiac or noncardiac problems)

☐ Drug overdose or poisoning with substance that causes Bradycardia, arrhythmia, QT prolongation, or QRS widening (eg, greater than 120 msec) (eg, beta-blockers, methadone, phenothiazines, sympathomimetic agents, cyclic antidepressants, digitalis, antiarrhythmic drugs)[AA]

- Short-term cardiac monitoring after therapeutic or diagnostic procedure requiring conscious sedation or anesthesia
- Acute cerebrovascular event[CC]
- Short-term cardiac monitoring during treatment with medications that increase risk of arrhythmia or Bradycardia (eg, lidocaine infusion for pain, neostigmine infusion for treatment of acute colonic pseudo-obstruction, phenytoin infusion for seizure without status epilepticus)

☐ Uncorrected electrolyte abnormalities associated with increased risk of dangerous arrhythmia (until corrected), as indicated by **1 or more** of the following:

- Hyperkalemia with attributable ECG changes
- Potassium less than 3.0 mEq/L (mmol/L)
- Potassium greater than or equal to 6.0 mEq/L (mmol/L) but less than 6.5 mEq/L (mmol/L) in patient with acute kidney injury, acute illness, or anticipated rapid rise in potassium
- Potassium 6.5 mEq/L (mmol/L) or greater

- Correction of severe metabolic acidosis or alkalosis
 - Frequent glucose checks (ie, more frequent than performable at lower level of care)
 - Severe fluid overload
 - Cerebral edema
 - Monitoring or suctioning for respiratory insufficiency or acidosis
 - Monitoring for active bleeding
 - Other need for treatment or monitoring not available outside the ICU
- ☐ **Systemic conditions**, including **1 or more** of the following:
- Severe electrolyte or metabolic disturbance causing or likely to cause **1 or more** of the following:
 - Life-threatening cardiac dysrhythmia
 - Respiratory insufficiency
 - Altered mental status
 - Seizures
 - Hemodynamic instability
 - Muscular weakness
 - Severe Hypothermia, as indicated by **1 or more** of the following:
 - Temperature 89.6 degrees F (32 degrees C) or lower
 - Temperature 95 degrees F (35 degrees C) or lower and severe organ dysfunction, as indicated by **1 or more** of the following:
 - Altered mental status
 - Hemodynamic instability
 - Respiratory depression
 - Pulmonary edema
 - Arrhythmia (eg, Bradycardia, atrial fibrillation, ventricular fibrillation)
 - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
 - Other new end organ dysfunction attributable to hypothermia
 - Environmental injuries such as hyperthermia, electrical injuries, or near drowning
 - Anaphylaxis with respiratory compromise, Hypotension, or other end organ dysfunction
 - Confirmed or suspected malignant hyperthermia as evidenced by exposure to volatile anesthetic agent or succinylcholine and **1 or more** of the following:
 - Hypermetabolism as evidenced by inappropriately increased CO₂ production, O₂ consumption, or acute mixed metabolic and respiratory acidosis
 - Unexplained Tachycardia

- Magnesium less than 1.3 mEq/L (0.65 mmol/L)
 - Magnesium greater than 6 mEq/L (3 mmol/L) or prolonged PR, QRS, or QT attributed to hypermagnesemia
 - Calcium[M] less than 8.6 mg/dL (2.15 mmol/L) or ionized calcium less than 4.65 mg/dL (1.16 mmol/L) with acute symptoms requiring intravenous repletion (eg, convulsions, severe muscle spasms, cardiac conduction disturbance)
 - Prolonged QT attributed to hypokalemia, hypomagnesemia, or hypocalcemia
 - Other clinically significant electrolyte abnormality associated with increased risk of arrhythmia
- ☐ Unexplained syncope or other neurologic event suspected of being due to arrhythmia or cardiac origin indicated by **1 or more** of the following:
- Hemodynamic instability
 - Outflow tract obstruction (eg, hypertrophic cardiomyopathy)
 - Structural abnormality (eg, certain congenital heart defects, ventricular dysfunction, cardiomyopathy)
 - Valvular disease (eg, aortic stenosis, prosthetic valve dysfunction)
 - Myocarditis
 - Patient reported palpitations or tachycardia preceding syncope
 - Syncope occurring during exercise
 - Syncope sudden and without prodrome
 - Syncope while supine
 - Suspected dysfunction of implantable pacemaker or defibrillator
 - History of Dangerous arrhythmia
 - History of previous syncope due to arrhythmia
 - Use of medication known to cause Dangerous arrhythmia
 - Family history of sudden death
 - Presentation consistent with acute coronary syndrome (eg, chest pain, cardiac ischemia)
 - Multiple syncopal episodes within last 6 months

- Cardiac tachyarrhythmias, ectopic ventricular beats or ventricular bigeminy
 - Masseter spasm
 - Muscle rigidity
 - Rapid increase in core body temperature
 - Acute rise in serum potassium, creatine kinase, or myoglobin
- Neuroleptic malignant syndrome as evidenced by exposure to neuroleptic drug (eg, phenothiazine, butyrophenone) or dopamine-depleting drug (eg, alpha-methyltyrosine), or withdrawal of dopaminergic agent (eg, levodopa, amantadine) and **1 or more** of the following:
 - Muscle rigidity
 - Hyperthermia
 - Altered mental status
 - Hemodynamic instability
 - Acute rise in creatine kinase
- Serotonin syndrome (serotonin toxicity) as evidenced by exposure to medication(s) that increases level of serotonin in CNS (eg, some antidepressants including serotonin reuptake inhibitors, monoamine oxidase inhibitors, opioids, stimulants, triptans, anticonvulsant agents, linezolid) and **1 or more** of the following:
 - Significant neurologic finding (eg, agitation, hypomania, hallucinations, Altered mental status, seizures)
 - Significant autonomic nervous system-related hyperactivity (eg, Hemodynamic instability, hypertension, sweating, hyperthermia, tachycardia, vomiting, diarrhea)
 - Significant musculoskeletal findings (eg, rigidity, myoclonus, hyperreflexia, tremor, rhabdomyolysis, elevated creatinine kinase)
- Severe alcohol withdrawal with **1 or more** of the following:
 - Hemodynamic instability
 - Cardiac arrhythmias of immediate concern
 - Uncontrolled seizures
 - Respiratory depression with need for, or high likelihood of requiring, mechanical ventilation
 - Need for anesthetic agent to control agitation (eg, dexmedetomidine, propofol, ketamine, intravenous phenobarbital)
 - Delirium tremens as evidenced by **ALL** of the following:
 - Cessation of, or reduction in, heavy and prolonged alcohol use
 - Delirium (eg, fluctuating disturbance in attention,
- Known channelopathy (eg, long QT syndrome, Brugada syndrome, or catecholaminergic paroxysmal ventricular tachycardia)
- High-risk ECG findings (eg, bifascicular block, Bradycardia, abnormal QT interval, ventricular pre-excitation, other conduction abnormality)

- awareness,
orientation, language)
- Signs or symptoms of alcohol withdrawal, as indicated by **2 or more** of the following:
 - Autonomic hyperactivity (diaphoresis, tachycardia, hypertension)
 - Tremor
 - Nausea or vomiting
 - Hallucinations
 - Increased anxiety
 - Psychomotor agitation
 - Generalized tonic-clonic seizures

☐ **General Surgery** diagnoses or procedures, including **1 or more** of the following:

- Acute abdominal catastrophe (eg, ischemic bowel, perforated viscus, abdominal compartment syndrome)
- Sepsis with end organ failure (eg, from abdominal abscess, infected gallbladder, or diverticulum)
- Complications of any surgery requiring ICU intervention, as indicated by **1 or more** of the following:
 - Hemodynamic instability
 - MI with complications (eg, severe arrhythmia, Hypotension)
 - Excessive bleeding or severe coagulopathy
 - Respiratory failure or insufficiency
 - Acute renal failure (stage 3 acute kidney injury)
 - Airway instability or obstruction
 - Neurologic deterioration
- Abdominal trauma with **1 or more** of the following:
 - Hemodynamic instability
 - Uncontrolled bleeding
 - Need for intubation and mechanical ventilation
 - High-grade splenic injury (eg, ruptured hematoma, splenic vascular injury)
 - Moderate to severe hepatic injury (American Association for the Surgery of Trauma (AAST) injury scoring scale grade III or higher, major hepatic vein injuries or avulsion)
 - Following catheter-directed embolization of abdominal organ bleed
 - Following damage-control surgery
 - Abdominal compartment syndrome requiring close monitoring or intervention, as indicated by **ALL** of the following:

- Sustained intra-abdominal pressure (eg, bladder pressure) greater than 20 mm Hg
 - New organ dysfunction or failure (eg, respiratory failure, Hemodynamic instability, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), intestinal ischemia) due to increased intra-abdominal pressure
- Multiple trauma with complicating features, as indicated by **1 or more** of the following^[P]:
 - Serious injury involving more than one organ or system
 - Single organ injury requiring critical care intervention or monitoring (eg, invasive hemodynamic monitoring, frequent vital signs or neurologic checks)
 - Impending acute respiratory failure due to lung contusion, unstable chest wall, aspiration, hemorrhage, tension or open pneumothorax, or fat embolism
 - Facial or neck injury threatening airway patency
 - Cardiac contusion
 - Pericardial effusion
 - Bronchial tear
 - Hemodynamic instability
 - Rhabdomyolysis requiring large-volume IV fluid resuscitation
 - Compartment syndrome
 - Other significant complicating feature
- Organ transplant
- Esophagectomy with **1 or more** of the following:
 - Need for postoperative mechanical ventilation
 - Hemodynamic instability
 - Postoperative complication requiring ICU intervention (eg, myocardial infarction, excessive bleeding, Acute renal failure (stage 3 acute kidney injury))
- Pancreatectomy (eg, pancreaticoduodenectomy, also known as Whipple procedure)^[Q]
- Necrotizing soft tissue infection (eg, necrotizing fasciitis, *Clostridial* gas gangrene, Fournier gangrene)
- Preoperative or postoperative patient requiring ICU intervention, such as hemodynamic optimization, pulmonary artery monitoring, mechanical ventilation, or extensive nursing care
- Obesity surgery patient with **1 or more** of the following:

- ICU management needs for comorbid conditions (eg, airway issues due to severe sleep apnea)
- Failed postoperative extubation
- Intraoperative complications (eg, perforated viscus, bleeding)

Hospitalization

Goal Length of Stay: 10 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Discharge Readiness

- Discharge readiness is indicated by patient meeting Recovery Milestones, including **ALL** of the following:
 - Hemodynamic stability
 - No evidence of biliary dysfunction
 - No evidence of ileus or bowel obstruction
 - Hypoxemia absent
 - Pain absent or managed
 - Pleural effusion absent or improved
 - No evidence of rejection
 - No evidence of postoperative or surgical site infection
 - Laboratory tests confirm adequate liver function and coagulation
 - Mental status at baseline
 - Electrolyte abnormalities absent or acceptable for next level of care
 - Renal function at baseline or acceptable for next level of care
 - Ambulatory or acceptable for next level of care^[DD]
 - Oral hydration^[EE]
 - Oral medications or regimen acceptable for next level of care
 - Oral diet or acceptable for next level of care
 - Discharge plans and education understood

See Liver Transplant, Pediatric Optimal Recovery Course [ISC](#) for full optimal recovery management information.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for:
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Continued Hypotension
 - Expect brief stay extension.
 - Hemorrhage, portal vein thrombosis, pancreatitis
 - Postoperative hemorrhage may require factor replacement, transfusion, and possible re-exploration.
 - Portal vein thrombosis may require anticoagulation, endovascular intervention, or surgical exploration.
 - Expect brief to moderate stay extension.
 - Biliary complications, leaks
 - Small bile leaks may heal spontaneously.
 - Anticipate imaging studies and ERCP with stenting or percutaneous transhepatic biliary drainage, or surgical intervention for large or nonresolving bile leak.
 - Expect moderate stay extension.
 - Continued metabolic abnormalities (eg, hyponatremia, hypernatremia, electrolyte disorder, hypervolemia)
 - Expect brief to moderate stay extension.
 - Damage control surgery if severe intraoperative difficulty or complications upon reperfusion (eg, requiring deferred completion of biliary reconstruction)
 - Expect moderate stay extension.
 - Acute hepatic necrosis
 - Expect brief to moderate stay extension.
 - Very severe preoperative liver disease
 - Higher preoperative Model for End-Stage Liver Disease (MELD/PELD) scores (eg, above 20) correlate with longer lengths of stay.
 - Anticipate brief to moderate stay extension.
 - Liver transplant performed in neonatal period (eg, secondary to congenital malformation)
 - Expect brief to moderate stay extension.
 - Acute rejection or graft dysfunction causing mild to moderate liver failure

- Expect brief to moderate stay extension.
- Hepatic artery thrombosis
 - Anticipate imaging studies and surgical or interventional radiology-guided revascularization, or possible retransplant.
 - Expect brief to moderate stay extension.
- Cardiac complications or dysfunction
 - Pulmonary edema, arrhythmia may occur.
 - Anticipate diuresis, antiarrhythmic agent, vasopressor agent, or possible cardiac intervention.
 - Expect brief to moderate stay extension.
- Pulmonary complications
 - Pleural effusion, atelectasis, acute respiratory distress syndrome, postoperative hepatopulmonary syndrome with Hypoxemia, portopulmonary hypertension, transfusion-related acute lung injury, or pneumonia may complicate recovery.
 - Anticipate thoracentesis, chest physiotherapy, possible pulmonary vasodilator or antibiotic therapy, and possible need for noninvasive or invasive mechanical ventilation.
 - Expect brief to moderate stay extension.
- Severe nutritional deficiency
 - Expect brief stay extension.
- Acute kidney injury or hepatorenal syndrome
 - Patients with worse preoperative synthetic function (eg, elevated INR) are at higher risk of postoperative acute kidney injury.
 - Expect brief stay extension.
- Infectious complications
 - Expect brief to moderate stay extension.
- Continued postoperative ileus
 - Expect brief stay extension.
- Altered mental status
 - Expect brief to moderate stay extension.
- ☐ Other complication or condition, including:
 - ☐ Anemia
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Active blood loss absent
 - Signs and symptoms of anemia absent or acceptable for next level of care
 - Mental status normal or at baseline
 - Hgb/Hct level stable and acceptable for next level of care
 - Etiology of anemia requiring inpatient care absent
 - ☐ Arrhythmia
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Arrhythmia evaluation completed
 - Reversible causes of arrhythmia eliminated or mitigated
 - Specific antiarrhythmia treatment initiated as appropriate
 - Anticoagulation requirements addressed (or anticoagulation not required). See Anticoagulation Requirement: Common Complications and Conditions [ISC](#) for further information.
 - ☐ Electrolyte disorder
 - Extended stay beyond goal length of stay for primary condition may be needed until **1 or more** of the following is present:
 - Electrolyte abnormality manageable at next level of care[FF]
 - No electrolyte abnormality, as indicated by **ALL** of the following:
 - Potassium greater than 3.5 mEq/L (mmol/L) and less than 5.0 mEq/L (mmol/L)
 - Sodium greater than 135 mEq/L (mmol/L) and less than 145 mEq/L (mmol/L)
 - Calcium greater than 8.5 mg/dL (2.13 mmol/L) and less than 10.5 mg/dL (2.63 mmol/L)
 - Phosphorus greater than 2.5 mg/dL (0.81 mmol/L) and less than 5.0 mg/dL (1.62 mmol/L)
 - Magnesium greater than 1.5 mEq/L (0.75 mmol/L) and less than 3.0 mEq/L (1.5 mmol/L)
 - ☐ Fever
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Hypoxemia absent
 - Tachypnea absent
 - ☐ Temperature status acceptable, as indicated by **1 or more** of the following:
 - Temperature less than 100.5 degrees F (38.1 degrees C) (oral)
 - Temperature as expected for disease process and acceptable for next level of care
 - Cultures negative or infection identified and under adequate treatment
 - No condition, organ dysfunction (eg, renal failure), or laboratory finding (eg, acidosis) identified that requires inpatient care
 - Behavior or mental status abnormalities absent or manageable at next level of care. See Mental Status Change: Common Complications and Conditions [ISC](#) for further information.
 - Adequate diet tolerated
 - Medical comorbidities absent or manageable at next level of care
 - ☐ Gastrointestinal bleeding
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Bleeding absent

- Stable Hgb/Hct
 - Platelet count, prothrombin time, and partial thromboplastin time acceptable for next level of care
 - Surgical or other acute intervention not needed
 - Mental status at baseline
 - Oral hydration and diet tolerated
- [-] Hyperglycemia and diabetes control
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Mental status at baseline
 - No precipitating cause identified, or cause manageable at next level of care
 - Acidosis absent
 - Blood glucose under acceptable control
 - Dehydration absent
 - Electrolyte abnormalities absent or acceptable for next level of care
 - Renal function at baseline or acceptable for next level of care
 - Outpatient diabetic medication regimen established
 - Oral hydration
 - Oral medications or regimen acceptable for next level of care
 - Oral diet or acceptable for next level of care
- [-] Mental status change
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Underlying medical etiology of mental status change absent, or manageable at next level of care
 - Danger to self or others absent or manageable at next level of care
 - Behavioral crisis management, including physical restraint or emergent pharmacologic treatment of agitation, not required or available at next level of care
 - Substance or alcohol withdrawal absent or manageable at next level of care
 - Behavioral symptoms (eg, agitation, somnolence, inappropriate behavior) absent or manageable at next level of care
- [-] Psychiatric disorders
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Danger to self or others absent or manageable at next level of care
 - Behavior crisis management, including physical or chemical restraints, not required or available at next level of care
 - Behavioral symptoms (eg, agitation, somnolence, inappropriate behavior) absent or manageable at next level of care
 - Functional status is sufficient to participate at available next level of care.
 - Patient and supports understand follow-up treatment and crisis plan.
 - Provider and supports are sufficiently available at next level of care.
 - Patient can participate (eg, verify absence of plan for harm) in needed monitoring.
- [-] Respiratory insufficiency
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Suctioning, pulmonary toilet, or other therapy not needed or available at next level of care
 - Respiratory insufficiency resolved or manageable at next level of care
 - Anticoagulation treatment not needed or available at next level of care
- [-] Urinary complications
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Mental status at baseline
 - Antibiotic regimen for next level of care established
 - Voiding adequately or with urinary catheter, percutaneous suprapubic tube, or plan for intermittent self-catheterization, and management regimen performable at next level of care
 - Urine output adequate
 - Renal function at baseline or acceptable for next level of care
 - Pain absent or managed
 - Oral intake adequate
 - Afebrile or temperature acceptable for next level of care
- [-] Venous thrombosis and pulmonary embolism
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability
 - Dyspnea absent
 - New oxygen requirement absent or supplementation can be performed at next level of care
 - Tachypnea absent
 - Dangerous arrhythmia absent
 - No bleeding or evidence of new emboli
 - Pain absent or managed
 - No evidence of cardiac dysfunction (eg, right ventricular heart failure)
 - Lower extremity examination at baseline, stable, or improved
 - Anticoagulation tolerated (eg, no allergic reaction)
 - Outpatient anticoagulation plan established
- [-] Wound and skin care
 - Extended stay beyond goal length of stay for primary condition may be needed until **ALL** of the following are present:
 - Hemodynamic stability

- Volume status acceptable (eg, not dehydrated)
- Mental status at baseline
- Hypoxemia absent
- Tachypnea absent
- Fistulas, tunneling, or infection absent or manageable at next level of care
- Surgical management requiring inpatient care absent
- Afebrile or temperature appropriate for next level of care
- Pain absent or managed
- Wound care needs manageable at next level of care
- Drain absent or drain care manageable at next level of care
- Wound hematoma or seroma absent or manageable at next level of care

See Common Complications and Conditions [ISC](#) for further information.

See Liver Transplant, Pediatric [ISC](#) Optimal Recovery Guideline for complete guideline and further supporting content such as Supplemental Medicare Criteria (ie, CMS' Two-Midnight Rule exceptions), Alternatives to Admission or Procedure, Optimal Recovery Course, and Evidence Summary.

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Transplant, Pediatric guideline in Home Care.

Footnotes

[A] The standard definition of acute liver failure in children is acute liver disease of less than 8 weeks' duration plus either an INR greater than 2, or an INR greater than 1.5 with any degree of hepatic encephalopathy. However, for purposes of liver transplant, the Organ Procurement and Transplantation Network defines acute liver failure (also known as "fulminant liver failure") as the onset of hepatic encephalopathy within 8 weeks of the first signs or symptoms of liver disease. [A in Context Link 1]

[B] Hepatic encephalopathy is graded by West Haven criteria for patients age 3 years and older, or by the Whitington Scale for patients younger than 3 years of age. A calculator for West Haven criteria is available at <https://www.mdcalc.com/hepatic-encephalopathy-grades-stages>. Whitington Scale criteria for early encephalopathy (stage I and II) include inconsolable crying, sleep reversal, and inattention to task; moderate encephalopathy (stage III) is characterized by stupor, combativeness, and somnolence with hyperreflexia; and late encephalopathy (stage IV) is characterized by coma (either arousable to painful stimuli (IVa) or with no response to pain (IVb)) with absent reflexes and decerebrate or decorticate posturing. [B in Context Link 1]

[C] The Model for End-Stage Liver Disease (MELD) score is used for liver allocation for patients 12 years of age and older; an individual's score can be calculated at the Organ Procurement and Transplantation Network website: <http://optn.transplant.hrsa.gov/converge/resources/MeldPeldCalculator.asp?index=98>. [C in Context Link 1]

[D] The Pediatric Model for End-Stage Liver Disease (PELD) score, a model similar to the Model for End-Stage Liver Disease (MELD), has been developed for children 11 years of age and younger. An individual's score can be calculated at the Organ Procurement and Transplantation Network website: <http://optn.transplant.hrsa.gov/converge/resources/MeldPeldCalculator.asp?index=99>. [D in Context Link 1]

[E] In cases where the calculated Model for End-Stage Liver Disease (MELD) or Pediatric Model for End-Stage Liver Disease (PELD) score does not reflect the urgent need for a liver transplant, an appeal may be made to the Regional Review Board for a score exception. Examples of conditions eligible for MELD or PELD score exclusions include hepatopulmonary syndrome, primary hyperoxaluria, metabolic disorders, hepatocellular carcinoma, and cystic fibrosis with FEV-1 less than 40% predicted. [E in Context Link 1, 2]

[F] Pediatric patients who have chronic liver disease with a Pediatric Model for End-Stage Liver Disease (PELD) score greater than 25 and have a need for mechanical ventilation or continuous renal replacement, ongoing gastrointestinal bleeding requiring 30 mL/kg or more of red cell transfusions within 24 hours, or a Glasgow coma score less than 10 may be registered as a status 1B for transplant, indicating an urgent need for liver transplant. [F in Context Link 1]

[G] Hepatopulmonary syndrome is characterized by hypoxemia due to intrapulmonary vascular dilation in the setting of portal hypertension or cirrhosis. [G in Context Link 1, 2]

[H] Portopulmonary hypertension with a mean pulmonary artery pressure of greater than 35 mm Hg is associated with a high mortality rate post liver transplant. [H in Context Link 1]

[I] Total hepatectomy with a temporary portacaval shunt may be required in severe hepatic trauma or primary liver allograft nonfunction immediately after transplant. [I in Context Link 1]

[J] Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture. [J in Context Link 1]

[K] The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals $1/3 \text{ SBP} + 2/3 \text{ DBP}$. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>. [K in Context Link 1, 2]

[L] Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia. Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated. Specific target values may require adjustment when care is delivered at high altitude. [L in Context Link 1]

[M] An ionized calcium level may be more accurate than a total serum calcium level, especially in the setting of acidosis, renal failure, multiple transfusions, hypoalbuminemia, or the presence of a paraprotein. [M in Context Link 1, 2, 3]

[N] A dedicated burn unit is recommended. [N in Context Link 1]

[O] Noninvasive ventilation may be administered in some intermediate care units depending on unit capabilities, staffing, and policies. [O in Context Link 1]

[P] Seriously injured patients may require transfer to a level 1 or level 2 trauma center. [P in Context Link 1]

[Q] Pancreaticoduodenectomy (also known as Whipple procedure) is a complex operation involving removal of the head of the pancreas, portions of the duodenum, gallbladder, and (sometimes) the stomach antrum, with anastomosis of the stomach to the jejunum, and reimplantation of the remaining body and tail of the pancreas and the common bile duct into a blind distal "Roux-en-Y" limb of the jejunum. Although the mortality of this procedure has greatly improved since its introduction, especially in high-volume centers, the procedure is still characterized by high morbidity. [Q in Context Link 1]

[R] Noninvasive ventilation may be administered in some intermediate care units depending on unit capabilities, staffing, and policies. [R in Context Link 1]

[S] A specialty guideline recommends the use of high-flow nasal oxygen over conventional oxygen therapy for patients with acute hypoxic respiratory failure because it decreases the risk of subsequent intubation and mechanical ventilation. High-flow oxygen is also recommended over conventional oxygen therapy for postextubation hypoxic respiratory failure due to a slightly lower likelihood for reintubation and improved patient comfort. [S in Context Link 1]

[T] For the purposes of the Inpatient & Surgical Care guidelines, telemetry care is defined as inpatient electrocardiographic monitoring not performed in an intensive or intermediate care unit. It is presumed that the patient meets clinical indications for admission or continued hospital stay, as the presence of a diagnosis or clinical situation listed in the Telemetry Care Guidelines does not necessarily indicate a need for inpatient care and monitoring. Telemetry care includes arrhythmia monitoring, and may include continuous ST-segment ischemia monitoring and monitoring for QTc prolongation. [T in Context Link 1]

[U] The continued need for telemetry care monitoring, including ST-segment monitoring and QTc prolongation monitoring, should be reassessed at least daily. Indications for telemetry care are often time-limited. Inappropriate use of telemetry care can be reduced by incorporating indication-based ordering into electronic ordering systems, excluding telemetry care from order sets when monitoring is not usually indicated, daily reviewing telemetry care orders and need for ongoing monitoring, using automated advisories suggesting discontinuation based on diagnosis and duration of telemetry, and educating the healthcare team on appropriate telemetry care indications. [U in Context Link 1]

[V] Telemetry care monitoring is not indicated in patients with Bradycardia who are hemodynamically stable and admitted for another reason. [V in Context Link 1]

[W] Telemetry care monitoring is not indicated in patients with chronic atrial fibrillation admitted for reason other than arrhythmia or rate, who are hemodynamically stable. [W in Context Link 1]

[X] Appropriate defibrillator detection and firing requires monitoring until treatment of inciting event. Inappropriate firing requires monitoring until the device is interrogated. [X in Context Link 1]

[Y] Telemetry care monitoring is not indicated for patients with a presumably functional defibrillator or pacemaker who are admitted for another reason. [Y in Context Link 1]

[Z] The initiation of certain antiarrhythmic medications may require continuous ECG monitoring because of proarrhythmic effects. [Z in Context Link 1]

[AA] Lists of agents associated with QT prolongation may be found at www.crediblemeds.org. [AA in Context Link 1, 2]

[BB] Routine uncomplicated cardiac procedures, such as coronary angiography, do not require cardiac monitoring once the femoral sheath is pulled post procedure. [BB in Context Link 1]

[CC] Dedicated intermediate-level stroke care units that incorporate rehabilitation are recommended over a telemetry care unit for acute ischemic stroke, if available. [CC in Context Link 1]

[DD] Patient is ambulatory or near baseline activity for age and development. [DD in Context Link 1]

[EE] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [EE in Context Link 1]

[FF] Patients with chronic disease (eg, renal failure with potassium greater than 5 mEq/L (mmol/L) or liver disease with sodium less than 135 mEq/L (mmol/L)) with chronically abnormal electrolytes are often manageable on an outpatient basis. [FF in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
 - Acute^[A] kidney injury (stage 2), as indicated by **1 or more** of the following^[B]:
 - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
 - Decreased urine output,^[C] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Urine output less than 0.5 mL/kg/hr for 12 hours
 - Significant clinical findings, as indicated by **1 or more** of the following^[B]:
 - Hemodynamic instability
 - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
 - Altered mental status
 - Cardiac arrhythmias of immediate concern
 - Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
 - Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
 - Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)^[D] or persists despite observation care
 - Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

References

- Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
- Weisbord SD, Palevsky PM. Prevention and management of acute kidney injury. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:940-977.e15.
- Palevsky PM, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for acute kidney injury. American Journal of Kidney Diseases 2013;61(5):649-72. DOI: 10.1053/j.ajkd.2013.02.349.
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Footnotes

- A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine. (1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
- ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute^[A] kidney injury (stage 3),^[B] as indicated by **1 or more** of the following^[C](1)(2)(3):
 - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
 - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m² (0.59 mL/sec/1.73m²)
 -  eGFR - Pediatric Calculator and **1 or more** of the following:
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
 - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
 - Decreased urine output,^[D] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Oligoanuria, as indicated by **1 or more** of the following:
 - Urine output less than 0.3 mL/kg/hour for 24 hours
 - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
 - Initiation of renal replacement therapy (RRT) required(4)

References

1. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
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4. Gaudry S, Palevsky PM, Dreyfuss D. Extracorporeal kidney-replacement therapy for acute kidney injury. New England Journal of Medicine 2022;386(10):964-975. DOI: 10.1056/NEJMra2104090.

Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
- C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
4. Schor NF. Neurologic evaluation. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:3053-3063.e1.

Altered mental status that is severe or persistent

- Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
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Bradycardia

- Bradycardia[A], as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rate less than 50 beats per minute in adult (age 18 years or older)[B]

- Heart rate less than 60 beats per minute in child or adolescent 6 to 17 years of age
- Heart rate less than 70 beats per minute in child 3 to 5 years of age
- Heart rate less than 80 beats per minute in child 1 to 2 years of age
- Heart rate less than 90 beats per minute in infant 6 to 11 months of age
- Heart rate less than 100 beats per minute in infant 3 to 5 months of age

References

1. Kusumoto FM, et al. 2018 ACC/AHA/HRS guideline on the evaluation and management of patients with bradycardia and cardiac conduction delay: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. *Circulation* 2019;140(8):e382-e482. DOI: 10.1161/CIR.0000000000000628. (Reaffirmed 2022 Jul)
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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. An evidence-based specialty society guideline has defined bradycardia in adults as less than 50 beats per minute.(1)

Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
 - Acute myocarditis
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
 - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
 - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
 - Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress
 - Association with other significant symptoms (eg, syncope)(9)(10)(12)

References

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Dangerous arrhythmia

- Dangerous arrhythmia, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Acute myocarditis
 - Myocardial ischemia
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress
 - Association with other significant symptoms (eg, syncope)(9)(10)(11)

References

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8. Park DS, Fishman GI. The cardiac conduction system. *Circulation* 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
9. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Journal of the American College of Cardiology* 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)
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11. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.

Dangerous arrhythmia absent

- Dangerous arrhythmia absent, as indicated by absence of **ALL** of the following(1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
 - Nonsustained ventricular tachycardia with myocarditis or ischemia
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Heart rhythms of concern due to association with Hypotension, Respiratory distress, or other significant symptoms (eg, syncope)

References

1. Curtis AB, Tomaselli GF. Approach to the patient with cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine*. 12th ed. Elsevier; 2022:1145-1162.e7.
2. Dalal AS, Van Hare GF. Disturbances of rate and rhythm of the heart. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. *Nelson Textbook of Pediatrics*. 21st ed. Philadelphia, PA: Elsevier; 2020:2436-2447.e1.
3. El-Chami M, Fernando L. Ventricular arrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1003-1014.

4. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. *Circulation* 2018;138(13):e272-e391. DOI: 10.1161/CIR.0000000000000549. (Reaffirmed 2023 Jul)
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7. Park DS, Fishman GI. The cardiac conduction system. *Circulation* 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
8. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Journal of the American College of Cardiology* 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)
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10. Westerman S, Manogue M, Hoskins MH, Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.
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Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
 - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
 - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
 - Hypotension that is severe, as indicated by **ALL** of the following:
 - Hypotension
 - Severe hypotension, as indicated by **1 or more** of the following:
 - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
 - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
 - Mean arterial pressure[A][D] less than 65 mm Hg
 - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
- C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated.(12)
- D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals $1/3 \text{ SBP} + 2/3 \text{ DBP}$.(1)(4) A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

1. Puskari MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
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5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾:
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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- Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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- Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Hypothermia

- Hypothermia indicated by core (eg, rectal) body temperature less than or equal to 95 degrees F (35 degrees C)⁽¹⁾

References

- Zafren K, Danzl DF. Hypothermia, frostbite, and nonfreezing cold injuries. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1750-1770.e2.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following⁽¹⁾:
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air^{[A][B]}
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)^[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

- Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
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Footnotes

- Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.⁽²⁾ Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.⁽²⁾ Specific target values may require adjustment when care is delivered at high altitude.⁽¹⁾⁽³⁾
- Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Hypoxemia absent

- Hypoxemia absent, as indicated by **1 or more** of the following(1):
 - Arterial oxygen saturation (SaO₂) of 90% or greater or arterial partial pressure of oxygen (PO₂) of 60 mm Hg (8.0 kPa) or greater on room air[A]
[B]
 - Oxygen requirements are stable and arranged for at next level of care.

References

- Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
- Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

- A. Specific target values (ie, 90% or 60 mm Hg (8.0 kPa)) may require adjustment when care is delivered at high altitude.(2)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

No contraindications to liver transplant present

- No contraindications to liver transplant present, as indicated by **ALL** of the following(1)(2)(3)(4)(5)(6)(7):
 - No sustained increases of intracranial pressure greater than 50 mm Hg
 - Cerebral perfusion pressure not less than 40 mm Hg for more than 2 hours
 - No irreversible brain damage
 - No advanced HIV disease[A](6)
 - No uncontrolled systemic infection (eg, cholangitis, bacterial peritonitis)
 - No active alcohol or illicit substance use[B]
 - No advanced cardiac or pulmonary disease (eg, untreated coronary artery stenosis of 70% or greater, severe valvular disease, severe chronic lung disease)
 - No unaddressed barriers to ability to comply with immunosuppression protocol and follow-up requirements (eg, adequate social support)
 - No history of persistent nonadherence
 - No anatomic abnormalities precluding liver transplant
 - No active extrahepatic malignancy[C]
 - No hemangiosarcoma
 - No Niemann-Pick disease type C

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Footnotes

- A. HIV-positive liver transplant candidates should have CD4 counts greater than 100/mm³ (0.1 x10⁹/L) and plasma HIV RNA levels that are expected to be undetectable at the time of liver transplant.(6)
- B. The duration of the period of sobriety required prior to liver transplant is controversial with no universally accepted time period. The most commonly employed duration is 6 months. Because studies have shown that mortality is high in patients with severe alcoholic hepatitis who do not undergo transplant, and that patients undergoing earlier liver transplant for severe alcoholic hepatitis have similar survival and alcohol relapse rates as those who have completed a 6-month waiting period, shorter durations of abstinence may be appropriate.(8)(9)(10)(11)(12) A specialty guideline recommends against basing the liver transplant candidacy on a fixed abstinence interval.(13)
- C. Liver transplant candidates with a history of extrahepatic malignancy should have received definitive treatment with an adequate tumor-free survival period. The determination of what constitutes an adequate tumor-free survival period varies depending on the type of malignancy and efficacy of the treatment received.(6)

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
 - Patient with **1 or more** of the following:
 - Dyspnea (difficulty breathing)
 - Tachypnea
 - Abnormal breathing pattern (eg, chest retractions)
 - Other evidence of difficulty breathing
 - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
 - Hypoxemia
 - Altered mental status
 - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following⁽¹⁾⁽²⁾:
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

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- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea^{[A][B]} as indicated by respiratory rate of **1 or more** of the following⁽²⁾⁽³⁾:
 - Greater than 18 breaths per minute in adult or child 13 years of age or older^[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age^[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age^[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age^[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age^[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age^[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age^[A]

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

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- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.⁽¹⁾ This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.⁽¹⁾

Tachypnea absent

- Tachypnea[A][B] absent, as indicated by respiratory rate of **1 or more** of the following⁽²⁾⁽³⁾:
 - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
 - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
 - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
 - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
 - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
 - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
 - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.
2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
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Codes

ICD-10 Diagnosis: C22.0, C22.2, C22.3, C22.4, C22.7, C22.8, C22.9, C78.7, C7B.02, E72.53, E74.00, E83.01, E83.110, E85.2, E88.01, K70.30, K70.31, K70.40, K70.41, K72.00, K72.01, K72.10, K72.11, K72.90, K72.91, K73.1, K73.2, K73.8, K74.3, K74.4, K74.5, K74.60, K74.69, K75.4, K75.81, K76.0, K76.2, K76.81, P78.81, Q44.2, Q44.3, Q44.5, Q44.6, T86.42 [Hide]

ICD-10 Procedure: 0FY00Z0, 0FY00Z1, 0FY00Z2

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